

# Sanjana Venkatesh

GitHub | LeetCode | HackerRank | Portfolio

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I'm someone who finds meaning in solving problems, asking the right questions, and turning uncertainty into something tangible. I approach challenges with curiosity, quiet determination, and a willingness to keep going until I find a solution that feels right.

## EDUCATION

### • RNS Institute of Technology

• *B.E. in Computer Science & Engineering*

Bengaluru, India

*Expected Jun 2027*

## EXPERIENCE

### • IEEE, KLE Section

• *Software Engineer Intern*

*Jun 2025 – Oct 2025*

- **Semantic Retrieval Backend:** Built a Node.js/Express backend matching natural-language physics questions to the correct simulation across a ~30-simulation corpus by generating Gemini embeddings, ranking via semantic similarity, and constraining a second Gemini call to the matched simulation's parameter schema for extraction.
- **API Robustness & Git Workflow:** Hardened the core API with structured 400/429/500 responses under a 10 req/60s limit via input validation, prompt gating, and CORS; reviewed PRs and led Git/GitHub workflow for a 4-member team.

### • Samsung Innovation Campus

• *IoT Intern*

*Sep 2025 – Jul 2026*

- **Embedded Systems Engineering:** Built an ESP32 touchscreen productivity dashboard (SPI TFT + capacitive touch) for real-time LeetCode/GitHub stat syncing without UI blocking, via non-blocking millisecond-timer scheduling with staggered Wi-Fi/HTTPS/GraphQL refresh calls; selected among 800+ applicants.

## PROJECTS

### • SimuPhysics | AI-Powered Physics Simulation Platform

Jun 2025 - Oct 2025

• *Node.js, Express, Gemini API, Matter.js, Vite*

*GitHub*

- Architected a full-stack app translating natural-language physics problems into interactive 2D simulations via a RAG pipeline: embedded scenario metadata with Gemini (**gemini-embedding-001**) and local cosine-similarity search to map prompts to 1 of 40+ simulations rendered with Matter.js.
- Leveraged Gemini's structured JSON schemas for parameter extraction from unstructured text, and optimized latency by running extraction plus markdown solution-generation concurrently via **Promise.all**.
- Implemented an AI-driven physics topic gate and IP-based rate limiting to protect LLM quota and ensure prompt relevance.

### • JanVaani | AI Government Circular Simplification Platform | (Part of First-Commit AWS Hackathon)

Sep 2026

• *FastAPI, React, MongoDB Atlas, AWS Bedrock/Polly/S3*

*GitHub*

- Built a full-stack platform (FastAPI + React/TypeScript + MongoDB) that ingests circulars from multiple government portals (PIB, DoPT, e-Gazette, state sources) via source-specific crawler adapters and simplifies them with AWS Bedrock summarization.
- Engineered multilingual translation (Hindi, Kannada, and other regional languages) and AWS Polly audio narration via chained translation/audio worker services for accessibility across literacy and visual-impairment barriers.
- Designed a human-in-the-loop admin dashboard (edit/approve/reject/redraft) backed by async ingestion/translation/audio workers, S3-backed provenance hashing, and a REST API separating public and bearer-token routes.

### • Nisa | Privacy-Focused Handheld Music Player | (Personal Project)

Aug 2026 - Ongoing

• *ESP32, RP2040, C++, LovyanGFX, LVGL*

*GitHub*

- Built a custom SPI driver for an ILI9341 TFT + XPT2046 resistive touch controller (LovyanGFX) with calibrated coordinate transforms (base-offset/delta-scaling, axis-swap) for pixel-accurate touch across portrait-mode rotations.
- Integrated LVGL (legacy v8.x and modern v9.x driver APIs) via custom display-flush/touchpad-read callbacks, raising touch-polling from the 30 Hz default to 100 Hz for responsive UI interaction.
- Shipped a placeholder UI (V1), then layered ESP32 Bluetooth connectivity on top across staged firmware branches, progressing from a bare display test to an interactive, wirelessly-connected device.

### • Sentinel Bootloader | Fault-Tolerant Bare-Metal Bootloader

March 2025 - Ongoing

• *C, ARM Cortex-M4, libopenm3, Make*

*GitHub*

- Designed a 3-strike, MRAM-persistent fault-triage system for an STM32F405RGT6 (Cortex-M4 @168MHz) that attributes watchdog-triggered resets to the last-active peripheral and power-gates it via the PMIC after 3 recorded faults, surviving single-event upsets/latch-ups across hard power cycles.
- Wrote a from-scratch boot sequence in C on libopenm3 (no HAL/RTOS, 2.5KB binary in a 32KB partition) that relocates the vector table (**SCB\_VTOR**) and branches directly to the flight OS at **0x08008000**.
- Armed a 1000ms hardware IWDG and persisted fault state to I2C-backed MRAM before every peripheral init, ensuring crash attribution independent of interrupt handlers; built a companion React/p5.js simulator to visualize MRAM/IWDG behavior pre-hardware.

## PROGRAMMING SKILLS

• **Languages:** Python, Java, JavaScript, C, C++, HTML5/CSS

**Backend & Systems:** Node.js, Express.js, FastAPI, Spring Boot, Django, REST APIs, Web Scraping (rate limiting, retries, parsing)

**Embedded & Firmware:** ARM Cortex-M4 (STM32), ESP32/RP2040, libopenm3, LVGL, LovyanGFX, Bare-metal C

**Databases:** PostgreSQL, MongoDB Atlas (cloud), SQLite, MySQLRelational & NoSQL Schema Design

**Cloud & Tools:** AWS (Bedrock, Polly, S3), Git/GitHub (PR workflows), Docker, JWT, Render, Vercel

**Problem Solving:** 400+ DSA (LeetCode, HackerRank); CodeChef rating 1386